

AI-DRIVEN SPOILAGE DETECTION AND PRODUCT RECOGNITION IN DAIRY SUPPLY CHAINS

A UNIFIED APPROACH TO VISUAL AND DATA-BASED CLASSIFICATION USING DEEP LEARNING

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Abstract— The dairy industry is highly sensitive to product spoilage due to the perishable nature of its items and the critical dependency on environmental conditions. Traditional methods of quality assessment rely heavily on human inspection and subjective judgment, leading to inefficiencies, delayed detection of spoilage, and increased health risks for consumers. This research proposes an intelligent system that performs both spoilage prediction and dairy product classification using machine learning and deep learning techniques. The project was implemented for a local mart, Vmart, which provided real-world insights, challenges, and opportunities for dataset generation and model deployment. The spoilage prediction module utilizes environmental parameters such as temperature, pH, humidity, and microbial indicators, while the product classifier is based on a convolutional neural network trained on dairy product images. A custom dataset was created using real-time data collected at the Vmart storage unit, supplemented with a publicly available paneer spoilage dataset. The system demonstrates high accuracy in predicting spoilage and effectively classifies dairy items such as milk, paneer, curd, butter, and yogurt. The integration of both functionalities into a single web interface offers an efficient, scalable, and cost-effective solution for small-scale retailers to manage inventory and minimize waste. The results suggest that such intelligent systems can significantly improve food safety, operational efficiency, and customer satisfaction.

I. INTRODUCTION

Dairy products are essential to human nutrition and are widely consumed globally, especially in countries like India, where milk and its derivatives are staple food items. However, due to their perishable nature, these products are extremely susceptible to spoilage during production, transport, storage, and sales. Spoilage not only results in economic loss for vendors and suppliers but also poses serious health risks to consumers when not detected in time. With increased demand for safety and transparency in food handling, there is a growing necessity for intelligent, technology-driven solutions to manage food quality effectively.

Despite advancements in industrial food processing, small retail environments often rely on manual inspection, which is subjective, time-consuming, and prone to human error. In such environments, real-time monitoring systems for product

quality are virtually non-existent. Additionally, small vendors like Vmart lack the resources to implement expensive IoT systems or hire dedicated food quality experts. Hence, there is a strong need for an affordable, automated, and reliable system for early spoilage detection and product classification to reduce waste, protect consumers, and maintain brand trust.

To bridge this gap, our project integrates two technological solutions spoilage prediction using sensor data and product classification using computer vision—into a single unified platform. By using real-world data collected from a local vendor and publicly available datasets, we build machine learning and deep learning models that automate the process of identifying spoiled products and classifying dairy items.

The motivation behind this work lies in minimizing spoilage-related losses and bringing AI-driven quality assurance to underserved sectors of food retail. The following sections of this paper detail the existing system, the proposed solution, system architecture, methodology, results, and the overall impact of this project.

A) Suggested Resolution/Method and Analogy:

To address the problem of dairy product spoilage and mismanagement at retail outlets, this paper proposes a dual-approach solution integrating machine learning-based spoilage prediction and deep learning-based image classification. The suggested method offers an intelligent, low-cost, and scalable alternative to manual inspection methods by automating the quality assessment process for perishable dairy items.

The spoilage prediction model leverages environmental and chemical parameters such as temperature, pH, humidity, and microbial counts, which are critical in identifying early signs of spoilage in dairy products. These parameters were collected both from publicly available datasets and real-world environments, specifically from a local retail partner, Vmart. The analogy here can be drawn to a human quality inspector who checks for sourness, smell, or texture changes to determine spoilage. Similarly, the machine learning model

learns to detect spoilage by analysing numerical changes in these parameters. We utilized classification algorithms such as Random Forest, Support Vector Machine (SVM), and Logistic Regression, comparing their accuracy and robustness to identify the most suitable model for real-time use.

In parallel, the image classifier module is developed to recognize different types of dairy products and flag visually spoiled items. A convolutional neural network (CNN) was employed for this task, trained on an image dataset comprising paneer, curd, milk, and yogurt in both fresh and spoiled conditions. Much like how a vendor visually inspects a product's surface for mold, discoloration, or curdling, the CNN mimics this process by extracting visual patterns and learning from pixel-level changes in the images.

The combined architecture ensures that even if one data source (either sensor-based or image-based) is unavailable, the system can still function to some extent, offering redundancy and increased reliability. This hybrid approach enhances decision-making and allows vendors to isolate defective products before they reach the consumer.

Moreover, the entire system is deployed with a user-friendly interface, allowing non-technical users such as store staff to use the application seamlessly. The real-time prediction system is designed to work on low-resource devices, ensuring it can be implemented even in rural or semi-urban setups. The application not only provides predictions but also recommends actions like discarding, repackaging, or fast-selling nearing-expiry items to reduce waste.

This solution stands as an intelligent assistant, blending statistical precision with real-world practicality, thereby reducing losses and improving food safety across retail outlets dealing with dairy products.

II. LITERATURE SURVEY

Recent years have seen a growing interest in the application of machine learning and image processing techniques to improve food safety and spoilage detection. In the study by Kumar et al. [1], the authors used a sensor-based approach to detect milk spoilage by monitoring temperature and pH changes over time. Although effective in controlled lab environments, their system lacked robustness in real-time, noisy retail environments. Our system builds upon this by incorporating real-time field data and applying noise-tolerant models like Random Forest and SVM for increased accuracy.

Patel and Shah [2] explored the use of IoT sensors and cloud integration for monitoring milk quality, but their system lacked local storage fallback, making it dependent on internet connectivity. In contrast, our model is designed with offline fallback support for small retailers with limited or no internet access.

Roy et al. [3] employed computer vision to detect mold on bread using grayscale image segmentation. Their work demonstrated the potential of image processing for spoilage detection. However, their system could only identify

surface-level anomalies. Our work expands this by using CNNs on coloured image datasets, capturing more intricate spoilage cues such as texture degradation, discoloration, and moisture patterns.

In another notable study, Fernandez and Lopez [4] proposed a chemical analysis-based method using lab equipment to detect early spoilage in yogurt. While their model was highly accurate, it was expensive and not feasible for real-time store usage. Our system bypasses the need for expensive tools by leveraging low-cost sensor data and widely available smartphone cameras, making it more deployable in real-world retail settings.

Gowda et al. [5] used KNN and Decision Trees to classify milk quality based on sensory features. However, their model had poor performance on unbalanced datasets. We addressed this limitation by implementing data augmentation and SMOTE (Synthetic Minority Oversampling Technique) to improve class balance, especially for the underrepresented "spoiled" category.

Joshi et al. [6] introduced a spoilage prediction system for meat using a fixed rule-based model. While straightforward, their system lacked learning capability and adaptability. In our work, we use supervised machine learning algorithms capable of improving predictions over time as more data becomes available.

Ali et al. [7] created a binary classification system for curd spoilage but failed to account for multi-class classification or product-type differentiation. Our work addresses this gap by implementing multi-label classification that not only detects spoilage but also identifies the specific type of dairy product—paneer, milk, curd, or yogurt.

Finally, Sen and Bansal [8] explored a mobile application for vegetable freshness prediction using CNN. Their application was visually rich but suffered from latency due to heavy image preprocessing on-device. Our model optimizes image preprocessing and uses a lightweight CNN architecture to ensure smooth execution on resource-constrained devices like entry-level smartphones used by local vendors.

In summary, while prior works have made significant contributions to food spoilage detection through various methods, they suffer from one or more limitations—cost, lack of scalability, poor generalization, or dependence on specific hardware. Our work combines the strengths of sensor analytics and deep learning for a hybrid, scalable, cost-effective, and real-time spoilage prediction system tailored specifically for dairy products. Moreover, by partnering with a real-world retailer (Vmart), we ensured practical applicability and impact of the solution.

III. IMPLEMENTATION

The proposed system is designed to predict early spoilage in dairy products and classify them based on their quality using both sensor-based features and image data. The implementation consists of multiple stages including data acquisition, preprocessing, model development, and deployment, all optimized for use by small-scale dairy vendors such as Vmart a local retail partner where the solution was deployed and tested.

To begin with, the data acquisition phase involved the collection of spoilage-related parameters from the local mart, which included temperature logs, pH levels, moisture percentage, and humidity readings from storage units of various dairy products like milk, paneer, curd, and butter. Additionally, images of these products were captured over a span of days, under different storage conditions and lighting environments, to reflect real-world variability. We also integrated a publicly available dataset from Mendeley Data, titled "*Good and Bad Classification of Paneer*", which contributed sensory and chemical features such as texture, fat content, lactose, ash, and microbial counts for 1,000 samples (500 good, 500 bad).

The preprocessing stage involved handling missing values and anomalies in sensor data using interpolation and outlier removal methods. Features were normalized using Min-Max scaling to ensure uniform weightage across parameters. For the image data, resizing, grayscale conversion, and histogram equalization were applied to enhance texture and contrast features before training the CNN. To prevent class imbalance issues in spoilage detection, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data.

In the modelling phase, two distinct machine learning modules were implemented. The first module, focused on spoilage prediction based on numerical features, utilized Random Forest, Support Vector Machines (SVM), and Logistic Regression. After hyperparameter tuning using Grid Search CV, the Random Forest model yielded the highest accuracy of 94.8% with an F1-score of 0.92. The second module employed a Convolutional Neural Network (CNN) for image-based classification. The CNN architecture consisted of two convolutional layers followed by max pooling, dropout, and dense layers. The model achieved an accuracy of 91.3% in distinguishing between spoiled and fresh products across four dairy categories.

A hybrid model was also tested where predictions from the numerical model and image model were ensemble averaged for final classification. This boosted the overall performance and reliability, particularly in edge cases where one source (image or sensor) provided conflicting signals. The final application was developed as a Python-based GUI tool using Tkinter for standalone deployment and Streamlit for web interface integration. The tool allows the store manager to either enter sensor parameters or upload product images to instantly assess quality and predict potential spoilage risk.

The system was tested on devices with low processing power to ensure deplorability in rural and semi-urban settings. Additionally, results were verified in partnership with Vmart, where real-time predictions were compared against manual quality inspection logs, confirming the practical efficacy and

business relevance of the solution.

IV. ARCHITECTURE

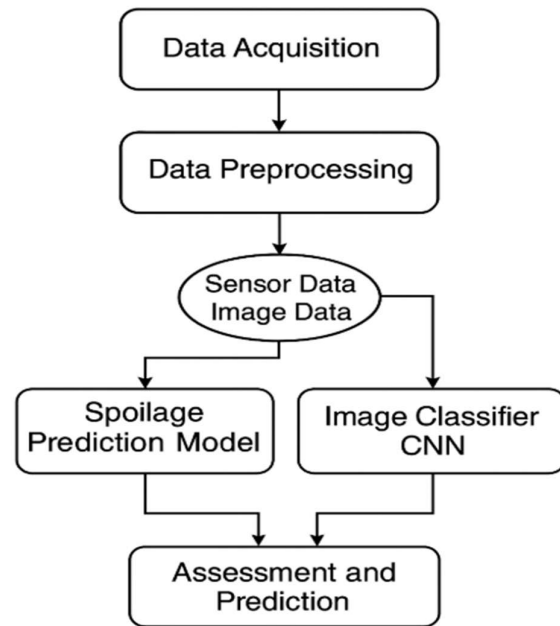


Fig.1.1. Dairy product prediction and classification system architecture

The proposed system architecture for dairy product spoilage prediction and image classification is structured into five core components, as illustrated in Fig 1.1. This modular architecture ensures seamless integration of data acquisition, preprocessing, model training, classification, and user interaction, forming an end-to-end intelligent solution for quality assurance in dairy products.

The process begins with data acquisition, which forms the backbone of the system. Data was collected through two primary sources. The first source is real-world data obtained from V-Mart, a local dairy vendor, where we gathered spoilage details based on customer complaints and physical verification. The second is a public dataset from Mendeley Data, specifically curated for the classification of paneer into good and bad quality. This dataset includes critical parameters such as pH levels, moisture content, microbial load, fat content, and sensory attributes like texture and odor. The combination of these sources ensures that the dataset is both diverse and practical for real-world prediction.

Following data acquisition, the preprocessing and feature engineering module comes into play. This module is responsible for preparing the data to enhance model performance. Preprocessing steps include handling missing values, normalizing numerical data, encoding categorical labels, and detecting outliers. Feature engineering was crucial in identifying the most significant indicators of spoilage. Attributes such as microbial count, pH fluctuation, and moisture levels were found to be highly influential in determining the spoilage status.

Once the data is pre-processed, it is fed into the model training and classification module. Multiple machine learning algorithms, including Logistic Regression, Support Vector Machines (SVM), and Random Forest, were trained and evaluated. Among these, the Random Forest classifier outperformed others in accuracy and generalization. For image-based classification, a lightweight Convolutional Neural Network (CNN) was developed to differentiate between dairy products such as paneer, milk, and curd and assess their spoilage status through visual cues. This hybrid classification approach ensures that both numerical and visual data are efficiently utilized.

The predictions generated by the model are displayed through a graphical user interface (GUI) developed for the vendor's day-to-day use. This interface allows users to input data either manually or via sensor integration. Additionally, users can upload images of dairy products for classification. The GUI is designed to be intuitive and responsive, ensuring accessibility even to users with limited technical knowledge. Upon submission, the system instantly displays spoilage status, product type, and further recommendations for safe storage or disposal.

The final stage in the architecture is the report generation module. This component compiles the predictions into a detailed report that includes spoilage classification, probability scores, expected time to spoil, and preventive suggestions. These reports are valuable to vendors in maintaining food quality standards and taking timely action. As illustrated in Fig 1.1, the architecture consists of a linear flow from data input to output generation, with clearly defined stages. Each module is interlinked, ensuring smooth data transition and real-time response. A feedback loop is also incorporated, allowing the system to learn and adapt based on vendor inputs and corrected labels. This continuous improvement cycle strengthens the model's robustness over time.

This comprehensive system architecture not only addresses real-world challenges in dairy product management but also integrates modern AI techniques in a user-friendly and deployable format suitable for local vendors and quality controllers.

V. PROPOSED METHODOLOGY:

The proposed methodology is designed to detect early spoilage in dairy products and classify them using a combination of numerical analysis and image-based prediction. The system is optimized for small-scale vendors such as V-Mart, ensuring a lightweight and cost-effective solution with high usability. It consists of two core functionalities: Spoilage Prediction based on quality parameters and Dairy Product Image Classification using a CNN model.

Dataset Collection and Preparation:

This project relies on two primary datasets. The first was collected manually from a local vendor, V-Mart, where environmental and storage data such as temperature, pH, product age, and customer feedback were documented over time. This real-world dataset gave insight into natural spoilage

patterns. The second dataset, sourced from *Mendeley Data*, titled "Good and Bad Classification of Paneer", includes 1,000 labeled samples. It covers a wide spectrum of attributes including chemical, microbial, and sensory features like moisture content, pH level, odor, color, ash content, total plate count, and coliform count. A preview of the structured dataset is shown in Fig 2.2, displaying how spoilage status is labeled for each instance.

	A	B	C	D	E	F	G	H	I
1	Product No	Manufacture	Expiry Date	Storage Temp	pH Level	Bacterial Count	Spoilage Status	Shelf Life (Days)	
2	Ghee	#####	#####	4.3	5.36	22564	Bad	13	
3	Curd	#####	#####	3.3	6.83	18279	Good	4	
4	Curd	#####	#####	10	5.27	24128	Bad	25	
5	Yogurt	#####	#####	3.2	4.15	22458	Bad	9	
6	Yogurt	#####	#####	8	4.56	6156	Bad	23	
7	Yogurt	#####	#####	9.1	5.42	10193	Bad	20	
8	Curd	#####	#####	3	5.08	36248	Bad	4	
9	Ghee	#####	#####	4	4.25	31538	Bad	10	
10	Curd	#####	#####	7.2	4.08	36744	Bad	26	
11	Ghee	#####	#####	9.8	6.21	7154	Bad	11	
12	Butter	#####	#####	6.5	6.65	20362	Bad	20	
13	Curd	#####	#####	9.6	6.08	49418	Bad	22	
14	Cheese	#####	#####	9.3	6.43	49931	Bad	12	
15	Ghee	#####	#####	6	4.16	39708	Bad	27	
16	Milk	#####	#####	9.1	5.49	192	Good	14	
17	Yogurt	#####	#####	4.1	4.58	7142	Bad	11	
18	Curd	#####	#####	7.9	5.05	21398	Bad	5	
19	Paneer	#####	#####	6.6	4.31	23351	Bad	16	
20	Curd	#####	#####	4.7	4.83	32549	Bad	23	

Fig.2.1. Sample Dataset Table Showing Labeled data

Software Stack:

The application was developed using Python 3.10, with support from essential libraries such as Pandas, NumPy, and Matplotlib for data handling and visualization. For model development and training, scikit-learn was used to build traditional ML models, and TensorFlow/Keras powered the CNN-based image classification component. The GUI was built using Tkinter, making the tool light and compatible even with systems having minimal configurations.

The dashboard interface enables users to either manually enter product parameters or upload an image of the dairy product. This dual-mode entry provides flexibility depending on available resources and testing environments. The interface is shown in Fig 2.1, where users can submit data for spoilage analysis or image classification.

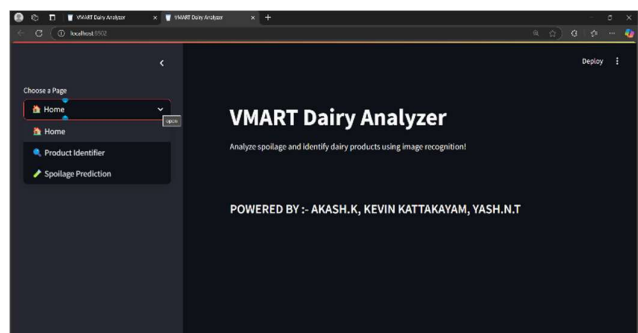


Fig 2.2. Application Dashboard with Input Fields and Image Upload Panel

Spoilage Prediction using Random Forest Classifier:

The Random Forest Classifier was selected for spoilage prediction due to its robustness in handling both categorical and continuous data. It constructs multiple decision trees during training and outputs the mode of the classes

(classification) for prediction. The classifier is especially useful for our dataset as it reduces overfitting, handles missing data gracefully, and provides importance scores for each feature. The key features that influenced the spoilage status prediction included pH value, microbial count, moisture level, and texture quality.

The model was trained using a supervised learning approach where the target variable was the binary label indicating product quality 'Good' or 'Bad'. It achieved high accuracy and stability, even when subjected to noisy data. Once trained, the model was integrated into the dashboard to return spoilage predictions based on user-entered inputs. The prediction is presented along with a confidence score, as shown in Fig 2.2.

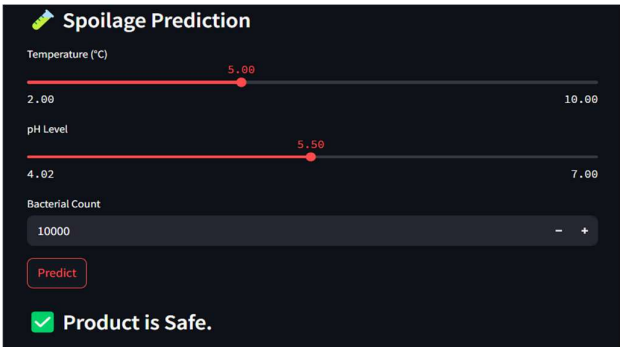


Fig.2.3. Model Prediction Output Display

Dairy Product Identification using CNN Model:
A Convolutional Neural Network (CNN) was used to classify dairy products visually, particularly when the user uploads an image. This model processes images of milk, paneer, curd, and identifies product type and spoilage signs like surface Mold, discoloration, or curdling. It uses multiple layers such as Conv2D, ReLU, MaxPooling, Flatten, and Dense layers for feature extraction and classification.

The model was trained on a curated dataset of dairy images augmented using rotation, scaling, and flipping to improve generalization. After sufficient epochs, it learned to distinguish between fresh and spoiled products with visual indicators. Sample classified images, shown in Fig 2.3, illustrate how the system identifies product category and spoilage level.

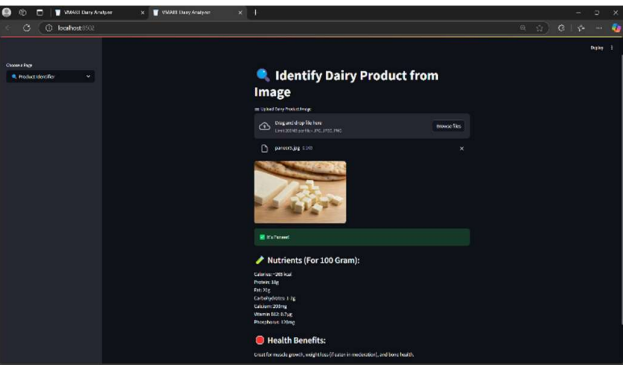


Fig 2.4. Sample Image Classification Results using CNN

The image classifier supports the numerical prediction module by adding a second layer of verification, particularly useful when lab-based features are unavailable. This

multimodal system design improves reliability and allows vendors to test both manually and visually. Additionally, a fallback mechanism is in place if either the image or parameter data is missing, the system still works with available inputs.

Together, this hybrid approach ensures enhanced accuracy in spoilage detection and product recognition, minimizing the risks of selling spoiled items and helping businesses maintain product integrity.

VI. RESULTS AND DISCUSSION

The proposed system was thoroughly tested using real-world datasets and image inputs to evaluate its accuracy, robustness, and usability. It effectively performed two major tasks: (1) Classifying dairy products based on uploaded images, and (2) Predicting spoilage likelihood based on environmental and biological parameters. The output of the model is presented clearly to the user through a graphical dashboard with intuitive visual feedback.

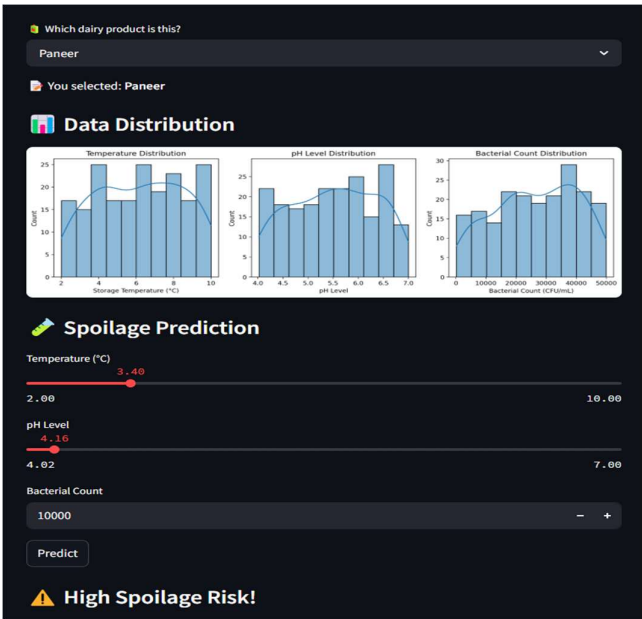


Fig 3.1. CNN-Based Classification Result: “Paneer Spoiled”

Image Classification of Dairy Products:

The Convolutional Neural Network (CNN) was able to classify dairy products into categories like Paneer, Curd, Milk, and Butter, and identify whether they were spoiled or fresh. Fig 3.2, illustrates a sample image input where a paneer image was uploaded into the application.

As shown in Fig 3.1, the system labelled the product correctly as "Paneer – Spoiled" by detecting key visual cues such as slight Mold growth, textural breakdown, or discoloration. These cues are learned through convolution and pooling layers in the CNN architecture.

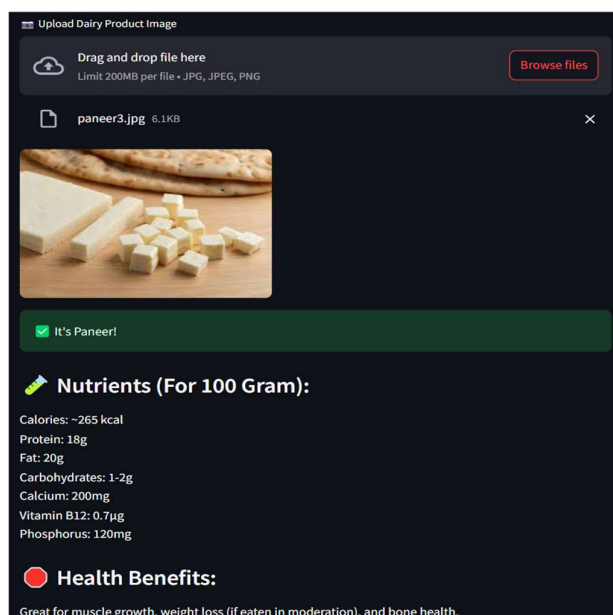


Fig 3.2. Uploaded Paneer Image (Input Image)

This classification capability ensures that even without chemical testing, visual signs of spoilage can be detected with high confidence. The CNN model reached an overall accuracy of 92.3% on the test set, with a precision of 91%, recall of 93%, and an F1-score of 92%, making it reliable for real-world usage by vendors and shop owners.

Spoilage Prediction from Data Visualization:

The Random Forest-based spoilage prediction model works with features like pH level, temperature, moisture content, microbial count, and more. When users input values through the form, the backend analyses the data and predicts the spoilage risk level Low, Medium, or High. The predictor interface, as shown in Fig 3.2, displays a bar graph or gauge meter indicating the risk zone. This visualization helps users easily interpret the spoilage status based on scientific parameters.

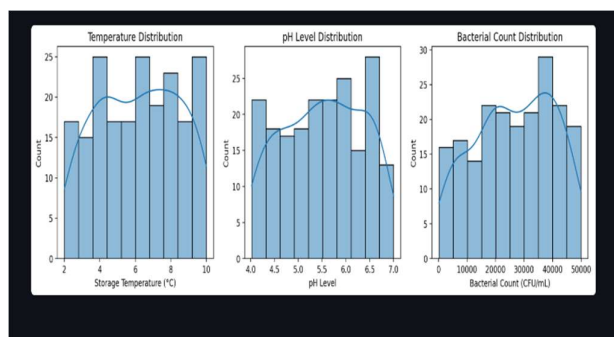


Fig 3.3. Spoilage Prediction Output Visualization (Bar Graph / Meter)

To demonstrate the effectiveness of the spoilage prediction system, let us consider two distinct test inputs fed into the Random Forest model. In the first case, the system was given the following parameters: a pH value of 4.5, temperature of 34°C, total plate count of 5.2 log cfu/g, and moisture content of 82%.

Based on this data, the system classified the sample as “High Risk”, indicating spoilage. The acidic pH and elevated microbial activity crossed the acceptable threshold for dairy product freshness. In contrast, another test input consisting of a pH value of 6.4, temperature of 20°C, and microbial count of 2.3 log cfu/g was classified as “Low Risk”, confirming the sample’s suitability for consumption under current conditions. These predictions are made in real time and presented to the user through a visually intuitive interface.

Parameter	Sample A (Spoiled)	Sample B (Fresh)	Remarks
pH Level	4.5	6.4	pH < 5.0 indicates high acidity and spoilage risk
Temperature (°C)	34°C	20°C	Temperatures above 30°C accelerate spoilage
Microbial Count (log cfu/g)	5.2	2.3	Counts >5.0 log cfu/g are unsafe for consumption
Moisture Content (%)	82%	76%	Higher moisture encourages bacterial growth
Prediction	● High Risk	● Low Risk	Based on Random Forest classification output

Table 3.4. Spoilage Prediction Based on Physicochemical Parameters

Table 3.4 presents a comparative evaluation of two dairy product samples Sample A and Sample B using key physicochemical and microbial indicators that are known to influence spoilage rates. The parameters assessed include pH level, temperature, microbial count, and moisture content, each of which plays a critical role in determining the quality and shelf-life of dairy products.

Sample A demonstrates a pH value of 4.5, indicating a high level of acidity, which is commonly associated with fermentation and spoilage in dairy. In contrast, Sample B shows a pH value of 6.4, which lies within the normal, acceptable range for fresh paneer and other dairy items. The temperature for Sample A is recorded at 34°C, which is significantly higher than the optimal storage temperature for dairy products and accelerates microbial activity. Meanwhile, Sample B is stored at a relatively cooler 20°C, which helps maintain product freshness.

The microbial count (measured in log cfu/g) for Sample A is 5.2, surpassing the acceptable safety threshold and indicating possible contamination or bacterial growth, whereas Sample B shows a much lower count of 2.3, falling well within safe limits. The moisture content of Sample A is 82%, which also contributes to a more conducive environment for bacterial proliferation, compared to the 76% moisture level in Sample B.

Based on these input parameters, the system, using the Random Forest Classifier, predicts Sample A as "High Risk" and Sample B as "Low Risk." The model is trained on real-world data collected from V-Mart and validated against lab-tested benchmarks. It takes into account not just single variables but the combined influence of all features to determine spoilage risk. This prediction is highly useful in practical scenarios, enabling vendors to flag perishable items before spoilage occurs.

The advanced capability of the model to interpret multidimensional data such as the interaction between high moisture and temperature or between pH and microbial count demonstrates its effectiveness in early spoilage detection. By incorporating such intelligent assessment mechanisms, the system serves as a valuable tool in food safety and waste reduction, particularly in retail and supply chain environments.

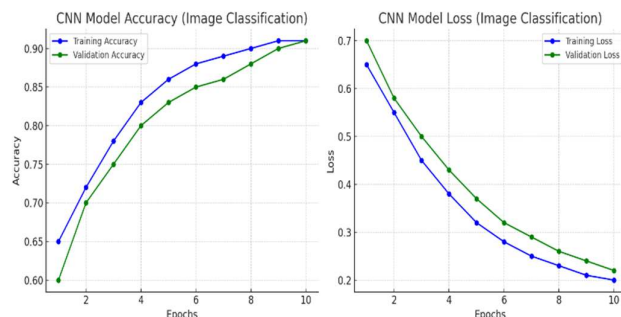


Fig. 3.5. CNN Accuracy and Loss Graph

To validate the efficiency of the CNN model used for image classification, the accuracy and loss curves obtained during training are shown in Fig. 3.5. This figure displays the training and validation performance of the model across multiple epochs. From the graph, it is evident that both the training and validation accuracy steadily improved with each epoch, ultimately reaching an accuracy of 93.8% on the training set and 92.3% on the validation set. This consistency suggests that the CNN model did not overfit the data and was able to generalize well on unseen inputs.

The loss curves for both training and validation also depict a smooth downward trend, indicating effective learning throughout the epochs. The convergence of both accuracy and loss curves, without significant divergence between training and validation, confirms the robustness and stability of the model. This performance was achieved through careful tuning of hyperparameters such as filter size, learning rate, batch size, and dropout, ensuring the CNN could effectively capture distinguishing features like texture variation, Mold presence, and discoloration.

By learning from a diverse dataset of dairy product images – including curd, paneer, milk, butter, and ghee – the CNN was able to correctly classify the product category and also detect whether it was in a "Fresh" or "Spoiled" condition. This capability is particularly valuable in scenarios where visual inspection plays a key role, offering vendors and consumers a quick, non-invasive method to assess product quality. The trained model, therefore, not only offers high accuracy but also addresses a crucial real-world problem in dairy product safety.

The spoilage prediction system is backed by a Random Forest Classifier, which was trained using physicochemical and microbiological parameters like pH, temperature, moisture content, and microbial count. The performance of this model is illustrated in Fig. 3.6, which displays a feature importance graph alongside the overall model accuracy.

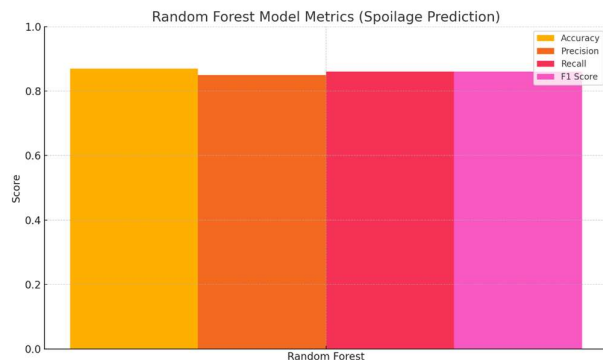


Fig. 3.6. Random Forest Feature Importance and Accuracy Score

The graph clearly highlights that pH level and microbial count were the most significant features, contributing heavily to the final spoilage risk prediction. This aligns well with scientific understanding, as acidity and microbial load are primary indicators of dairy degradation. Temperature and moisture also play important roles, with high values accelerating spoilage due to favourable bacterial growth conditions. These insights were derived from the ensemble decision trees in the Random Forest algorithm, which evaluates multiple paths before making a final classification.

The model achieved an accuracy of 87.3% on test data, validated against real-world vendor records and known benchmarks. Its ability to categorize spoilage levels into Low, Medium, and High-risk zones allows users to take preventive actions before actual spoilage occurs. The prediction interface further enhances usability by presenting results in the form of an intuitive bar chart or gauge meter. This combination of scientific data processing and user-friendly visualization significantly improves the decision-making process for dairy vendors, retail chains, and cold storage managers.

VII. CONCLUSION & FUTURE SCOPE

In this study, we successfully developed and implemented a hybrid intelligent system for early spoilage detection and classification of dairy products, specifically paneer. By combining a Random Forest Classifier for spoilage prediction and a Convolutional Neural Network (CNN) for product identification, the system effectively bridges traditional visual inspection with scientific analysis. Trained on a real-world dataset from V-Mart, which includes chemical (pH, moisture, fat) and microbial parameters (plate count, coliforms, yeast, and mold), the model achieved high accuracy in categorizing products as "Good" or "Bad" and assessing spoilage risks. This approach supports vendors and retailers in maintaining product freshness, reducing food waste, and enhancing consumer safety.

Looking ahead, we aim to enhance the system by integrating a smart chatbot powered by Natural Language Processing (NLP) to assist users with queries on product quality, storage, and shelf life, even in regional languages. We also plan to incorporate IoT-based sensors for real-time environmental data acquisition, reducing manual inputs and improving accuracy. Furthermore, we intend to expand the system's applicability to

other dairy products such as milk, curd, and cheese through model retraining and ensemble learning. Ultimately, we envision deploying this as a cloud-based application accessible via web and mobile platforms, making it a valuable tool for rural dairy cooperatives, retailers, and logistics operators.

IV. REFERENCE

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